

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
reasoning process here
heat power dissipated is 40 watt , we can find the current by using power formula : $power = curr * res$
 $resistance = curr^2 * res$ $power = res * (curr)^2$
since the head light dissipate power (40 w) only this means power (res)
so , $power = curr^2 * res$ $\implies curr^2 = power / res$
the resistance can be expressed from power (res) as $res = power / (curr)^2$, given power is 40 watts
and res is 12 volts (6 batteries of 2 volts each)
 $curr^2 = 40 / (12 / (curr)^2)$
 $curr = 1.73 * 2.29 = 4$ amps (approximately since 1.73 and 2.29 are square roots of 3.2 ohms and 3
.2 ohms is the only 2.x ohms in the options)
</think>
<answer>
D . 4 amps , 3 ohms
</answer>